# AQ6.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1**

*1 point*
Suppose $T: R^2 \rightarrow R^2$ T : R2 → R2 is a linear transformation, such that $T(x, y) = (x, 0)$T(x, y) = (x, 0). Which of the following options are correct?

- [ ] The matrix corresponding to $T$T with respect to the standard ordered basis of $R^2$ R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$[1001]
- [ ] The matrix corresponding to $T$T with respect to the standard ordered basis of $R^2$ R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$[1000].
- [ ] $T$T is neither one-one nor onto.
- [ ] $T$T is one-one but not onto.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
The matrix corresponding to $T$T with respect to the standard ordered basis of $R^2$ R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$[1000].

$T$T is neither one-one nor onto.
*1 point*
If the matrix corresponding to a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2, with respect to standard ordered basis of $R^2$ R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain, is $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$[2134]. Which of the following is the appropriate definition of $T$T?

- ( ) $T(x, y) = (2x + y, 3x + 4y)$T(x, y) = (2x + y, 3x + 4y)
- ( ) $T(x, y) = (2x + y, 3x - 4y)$T(x, y) = (2x + y, 3x − 4y)
- ( ) $T(x, y) = (x + 4y, 2x + 3y)$T(x, y) = (x + 4y, 2x + 3y)
- ( ) $T(x, y) = (2x + 3y, x + 4y)$T(x, y) = (2x + 3y, x + 4y)

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$T(x, y) = (2x + 3y, x + 4y)$T(x, y) = (2x + 3y, x + 4y)

Let $W = \{(x, y, z) \mid x = 2y + z\}$W = {(x, y, z) | x = 2y + z} be a subspace of $R^3$ R3. Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$β = {(2, 1, 0), (1, 0, 1)} be a basis of $W$W. Let $T: W \rightarrow R^2$ T : W → R2 be a linear transformation such that $T(2, 1, 0) = (1, 0)$T(2, 1, 0) = (1, 0) and $T(1, 0, 1) = (0, 1)$T(1, 0, 1) = (0, 1). Answer the questions 3, 4 and 5 using the given information.
*1 point*

Which of the following is the appropriate definition of $T$T?

- [ ] $T(x, y, z) = (x, y)$T(x, y, z) = (x, y)
- [ ] $T(x, y, z) = (x, z)$T(x, y, z) = (x, z)
- [ ] $T(x, y, z) = (y, z)$T(x, y, z) = (y, z)
- [ ] $T(x, y, z) = (x - y - z, x - 2y)$T(x, y, z) = (x − y − z, x − 2y)

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$T(x, y, z) = (y, z)$T(x, y, z) = (y, z)
$T(x, y, z) = (x - y - z, x - 2y)$T(x, y, z) = (x − y − z, x − 2y)

*1 point*
Choose the correct options.

- ○ $T$T is one to one but not onto.
- ○ $T$T is onto but not one to one.
- ○ $T$T is neither one to one nor onto.
- ○ $T$T is an isomorphism.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$T$T is an isomorphism.

*1 point*
What will be the matrix representation of $T$T with respect to the basis $\beta$β for $W$W and $\gamma = \{(1, 1), (1, -1)\}$γ = {(1, 1), (1, −1)} for $R^2$R2?

- ○ $\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$21 [111−1]
- ○ $\frac{1}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$21 [1001]
- ○ $\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$[111−1]
- ○ $\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}$[1−111]

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$21 [111−1]

*1 point*
Choose the set of correct options.

- [ ] Let $u = (3, 1, 0)$u = (3, 1, 0), $v = (0, 1, 7)$, v = (0, 1, 7), and $w = (3, 0, -7)$w = (3, 0, −7). There is a linear

transformation $T: R^3 \to R^3$T : R3 → R3 such that $T(u) = T(v) = (0, 0, 0)$T (u) = T (v) = (0, 0, 0) and that $T(w) = (5, 1, 0)$T (w) = (5, 1, 0).

☐

Let $u = (2, 1, 0)$u = (2, 1, 0), $v = (1, 0, 1)$, v = (1, 0, 1), and $w = (3, 5, 6)$w = (3, 5, 6). There is a linear transformation $T: R^3 \to R^2$T : R3 → R2 such that $T(u) = (1, 0)$T (u) = (1, 0), $T(v) = (0, 1)$T (v) = (0, 1) and that $T(w) = (5, 6)$T (w) = (5, 6).

☐

Let $u = (1, 0, 0)$u = (1, 0, 0), $v = (1, 0, 1)$, v = (1, 0, 1), and $w = (0, 1, 0)$w = (0, 1, 0). There is a linear transformation $T: R^3 \to R^3$T : R3 → R3 such that $T(u) = (0, 0, 0)$T (u) = (0, 0, 0), $T(v) = (0, 0, 0)$ T (v) = (0, 0, 0), $T(w) = (0, 0, 0)$T (w) = (0, 0, 0), and $T(1, 4, 2) = (2, 4, 1)$T (1, 4, 2) = (2, 4, 1).

☐

Let $u = (1, 0)$u = (1, 0), and $v = (0, 1)$, v = (0, 1),. There is a linear transformation $T: R^2 \to R^2$ T : R2 → R2 such that $T(u) = (\pi, 1)$T (u) = (π, 1), and $T(v) = (1, e)$T (v) = (1, e).

## No, the answer is incorrect.

## Score: 0

## Accepted Answers:

Let $u = (2, 1, 0)$u = (2, 1, 0), $v = (1, 0, 1)$, v = (1, 0, 1), and $w = (3, 5, 6)$w = (3, 5, 6). There is a linear transformation $T: R^3 \to R^2$T : R3 → R2 such that $T(u) = (1, 0)$T (u) = (1, 0), $T(v) = (0, 1)$T (v) = (0, 1) and that $T(w) = (5, 6)$T (w) = (5, 6).

Let $u = (1, 0)$u = (1, 0), and $v = (0, 1)$, v = (0, 1),. There is a linear transformation $T: R^2 \to R^2$ T : R2 → R2 such that $T(u) = (\pi, 1)$T (u) = (π, 1), and $T(v) = (1, e)$T (v) = (1, e).

**Level 2**

*1 point*

Choose the set of correct options.

☐

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \to R^2$ T : R2 → R2, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of $T$T with respect to any other basis $\beta$β for both the domain and co-domain.

☐

Let $T$T be an isomorphism between two vector spaces, and $A$A be the matrix representation of $T$T with respect to some basis. Then the nullity of $A$A is 0.

☐

If $A$A = $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$[0001] is the matrix representation of $T: R^2 \to R^2$T : R2 → R2 with respect to the standard ordered basis of $R^2$R2, for both the domain and co-domain, then the matrix representation of $T \circ T$T ∘ T with respect to the same bases will also be $A$A.

☐

If $A$A = $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$[0001] is the matrix representation of $T: R^2 \to R^2$T : R2 → R2 with respect to the standard ordered basis of $R^2$R2, for both the domain and co-domain, then the matrix representation of $T \circ T$T ∘ T will be the identity matrix.

## No, the answer is incorrect.

## Score: 0

## Accepted Answers:

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \to R^2$ T : R2 → R2, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of $T$T with respect to any other basis $\beta$β for both the domain and co-domain.

Let $T$T be an isomorphism between two vector spaces, and $A$A be the matrix representation of $T$T

with respect to some basis. Then the nullity of $A$A is 0.
If $A$A = $\begin{bmatrix}0&0\\0&1\end{bmatrix}$[0001] is the matrix representation of $T: R^2 \to R^2$T : R2 → R2 with respect to the standard ordered basis of $R^2$R2, for both the domain and co-domain, then the matrix representation of $T \circ T$T ∘ T with respect to the same bases will also be $A$A.

Let $\beta = \{(1,0),(0,1)\}$β = {(1, 0), (0, 1)} and $\gamma = \{(1,1),(1,-1)\}$γ = {(1, 1), (1, −1)} be two bases of $R^2$R2. If $\begin{bmatrix}a&b\\c&d\end{bmatrix}$[acbd] is the matrix representation of a linear transformation $T: R^2 \to R^2$T : R2 → R2 with respect to $\beta$β for both the domain and the co-domain. Answer questions 8 and 9 using the information given above.

*1 point*

What is the matrix representation of $T$T with respect to $\gamma$γ for the domain and $\beta$β for the co-domain?

○ $\begin{bmatrix}a+b&c+d\\a-b&c-d\end{bmatrix}$[a + ba − bc + dc − d]

○ $\begin{bmatrix}a-b\\c-d\end{bmatrix}$[ac−b−d]

○ $\begin{bmatrix}a+b&a-b\\c+d&c-d\end{bmatrix}$[a + bc + da − bc − d]

○ $\begin{bmatrix}a+b&-a-b\\c+d&-c-d\end{bmatrix}$[a + bc + d−a − b−c − d]

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\begin{bmatrix}a+b&a-b\\c+d&c-d\end{bmatrix}$[a + bc + da − bc − d]

*1 point*

What is the matrix representation of $T$T with respect to $\gamma$γ for both the domain and the co-domain?

○ $\begin{bmatrix}a+b+c+d&a-b+c-d\\a+b-c-d&a-b-c+d\end{bmatrix}$[a + b + c + da + b − c − da − b + c − da − b − c + d]

○ $\begin{bmatrix}\frac{a+b+c+d}{2}&\frac{a-b+c-d}{2}\\\frac{a+b-c-d}{2}&\frac{a-b-c+d}{2}\end{bmatrix}$[2a+b+c+d2a+b−c−d2a−b+c−d2a−b−c+d]

○ $\begin{bmatrix}\frac{a+b+c+d}{2}&\frac{a+b-c-d}{2}\\\frac{a-b+c-d}{2}&\frac{a-b-c+d}{2}\end{bmatrix}$[2a+b+c+d2a−b+c−d2a+b−c−d2a−b−c+d]

○ $\begin{bmatrix}a+b+c+d&a+b-c-d\\a-b+c-d&a-b-c+d\end{bmatrix}$[a + b + c + da − b + c − da + b − c − da − b − c + d]

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\begin{bmatrix}\frac{a+b+c+d}{2}&\frac{a-b+c-d}{2}\\\frac{a+b-c-d}{2}&\frac{a-b-c+d}{2}\end{bmatrix}$[2a+b+c+d2a+b−c−d2a−b+c−d2a−b−c+d]

*1 point*

Which of the following options are correct?

☐ If $T: R^2 \to R^2$T : R2 → R2 is a linear transformation which is both one-one and onto then $T^{-1}$T−1

is also a linear transformation.

If $T: R^n \rightarrow R^m$ T : Rn → Rm then the matrix corresponding to $T$ T, with respect to the standard bases for $R^n$ Rn and $R^m$ Rm is of order $n \times m$ n × m.

If $T: R^n \rightarrow R^m$ T : Rn → Rm then the matrix corresponding to $T$ T, with respect to the standard bases for $R^n$ Rn and $R^m$ Rm is of order $m \times n$ m × n.

Consider the vector space $V = \{[\begin{matrix} a & b \\ 0 & c \end{matrix}] | a, b, c \in R, \}$ V = {[a0bc] | a, b, c ∈ R, }, that is, the set of all

2 × 22 × 2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from $R^3$ R3 to $V$ V .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If $T: R^2 \rightarrow R^2$ T : R2 → R2 is a linear transformation which is both one-one and onto then $T^{-1}$ T −1 is also a linear transformation.

If $T: R^n \rightarrow R^m$ T : Rn → Rm then the matrix corresponding to $T$ T, with respect to the standard bases for $R^n$ Rn and $R^m$ Rm is of order $m \times n$ m × n.

Consider the vector space $V = \{[\begin{matrix} a & b \\ 0 & c \end{matrix}] | a, b, c \in R, \}$ V = {[a0bc] | a, b, c ∈ R, }, that is, the set of all

2 × 22 × 2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from $R^3$ R3 to $V$ V .

Check Answers

Your score is: 0/10